Bioinformatics Quality Management System

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Objectives

- Describe quality management system (QMS) components
  - Reproducibility
  - Version control
  - Documentation

- Apply QMS principles to NGS workflows
  - Pre-analytical
  - Analytical

- Common troubleshooting

Reproducibility

- Same input + same methods = same results
  - Core principle of computational (all) science

- Record software versions
  - Aligners, samtools, variant callers, workflow managers
- Track changes with version control
  - Git for scripts and pipeline development
- Document parameters and references
  - Primer schemes
  - Quality thresholds

- Identify test data to run through new versions and pipelines

- Use workflow managers
  - Snakemake, Nextflow for structured, repeatable runs
- Leverage containers
  - Docker or Singularity ensure consistent environments

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Documentation

- Explains how tools and pipelines work
  - Inputs, outputs, parameters, and assumptions
- Enables reproducibility
  - Others (and future you) can rerun the same analysis
- Reduces errors and misuse
  - Clear defaults and examples prevent incorrect runs
- Essential for collaboration
  - Shared understanding across labs, teams, and institutions

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Best Practices in writing documentation

- README.md files
  - Overview, requirements, and basic usage
- Usage examples & command snippets
  - Copy-pasteable starting points
- Parameter descriptions
  - What flags do, expected values, and defaults
- Workflow diagrams
  - Visualize steps, inputs, and outputs

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Best Practices in writing documentation

- Document data formats
    - FASTQ, FASTA, BAM, reference versions
- Record software versions
    - Tools, containers, workflow managers
- Include example datasets
    - Small test cases to validate setup

Best Practices in using documentation

- Start with the README
  - Understand the purpose, inputs, outputs, and limitations
- Check versions and dates
  - Ensure the documentation matches the tool or pipeline version you're using
- Follow the example first
  - Run the provided test or example before real data
- Read parameter descriptions carefully
  - Defaults may not match your experiment
- Verify input formats
  - FASTQ, FASTA, BAM, sample sheets, reference builds
- Look for expected outputs
  - Confirm files and directories match the documented results

Git

- Git is a version control system
  - Tracks changes to files over time
- Designed for collaboration
  - Multiple people can work on the same codebase
  - Think of it like a shared document (Microsoft office, Google docs)
- Keeps a history of your work
  - See what changed, when, why, and by whom
- Common in bioinformatics
  - Pipelines, scripts, documentation, and workflows

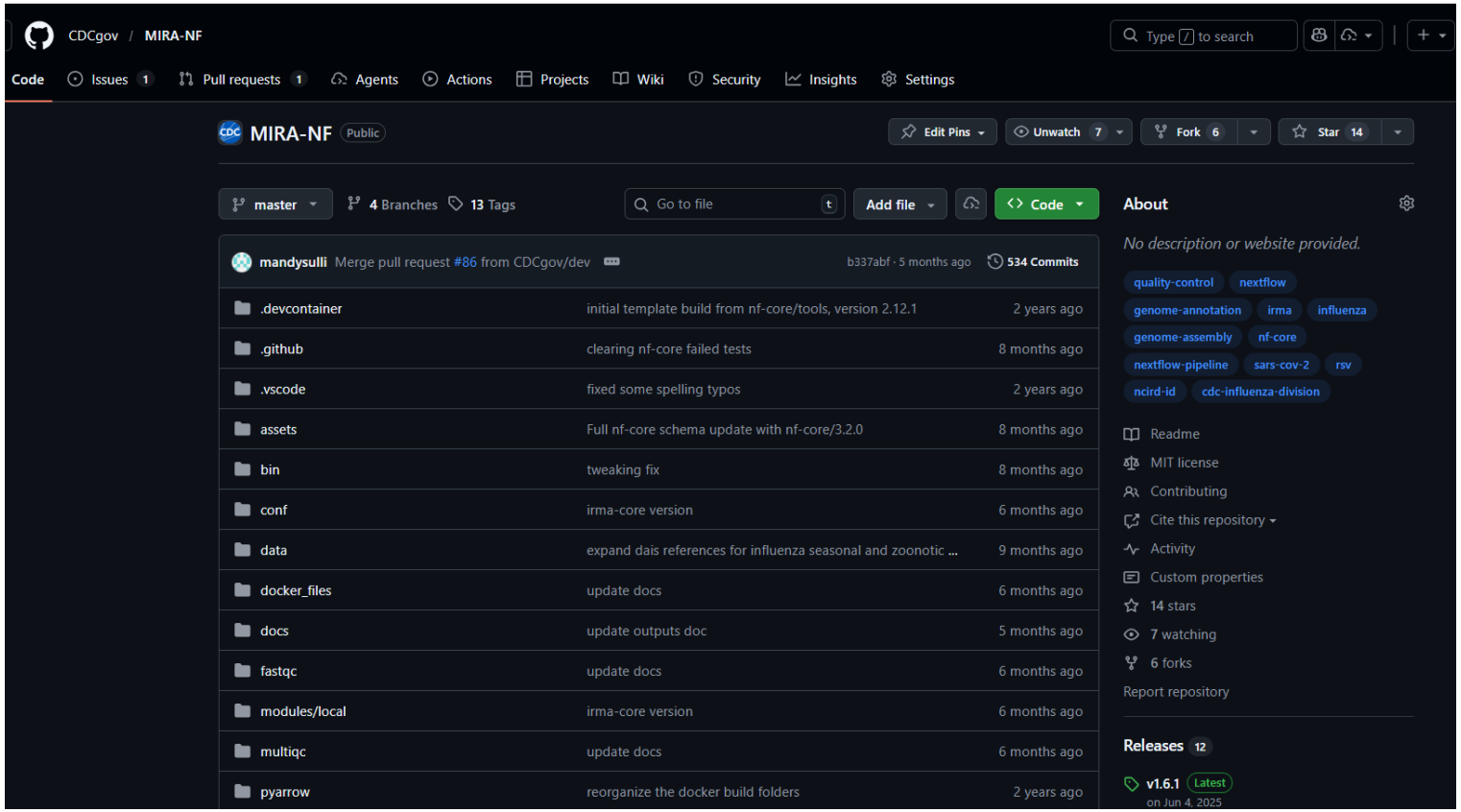
- Gitlab and Github most used

GitLab

GitHub

Core Git concepts

- Repository (repo)
  - A project tracked by Git
- Commit
  - A saved snapshot of changes with a message
- Branch
  - A parallel version of the code for development or testing
- Remote repository
  - Shared copy on platforms like GitHub or GitLab



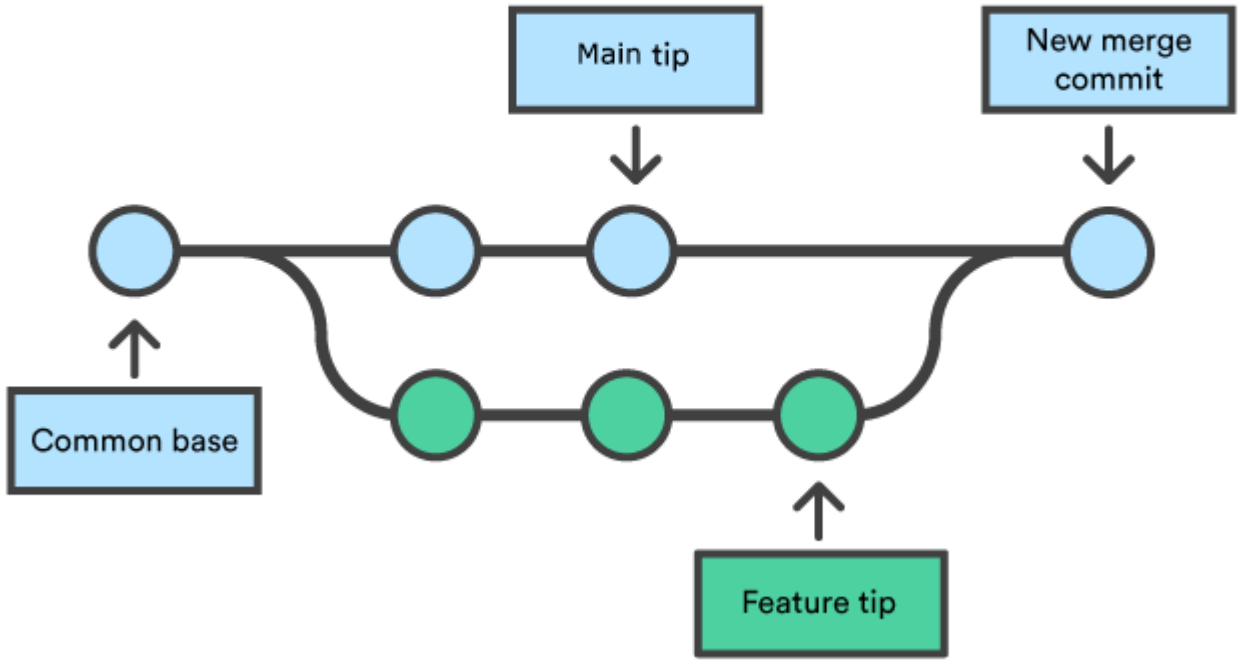

```
qgx6@rosalind02:MIRA-NF$ git status
On branch dev
Your branch is up to date with 'origin/dev'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .MIRA_nextflow.sh

nothing added to commit but untracked files present (use "git add" to track)
```

Git

- Common Git Commands (User Level)
- Create or get a repository
  - git init — start a repo
  - git clone <repo> — copy an existing repo
- Track and save changes
  - git status — see changes
  - git add <file> — stage changes
  - git commit -m "message" — save snapshot
- Sync with others
  - git pull — get updates
  - git push — share your changes



Common problems in Influenza bioinformatics

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Common issues

- Failed Mira QC
  - Low-coverage / Incomplete segment coverage
    - Not enough reads in your library → Back to the lab
      - Ct <= 28?
      - Gel image resolves segment bands cleanly?
      - Proper sample number on your flow cell?
    - MIRA <= v2.0.0 ??
      - Reads are being subsampled

Common issues

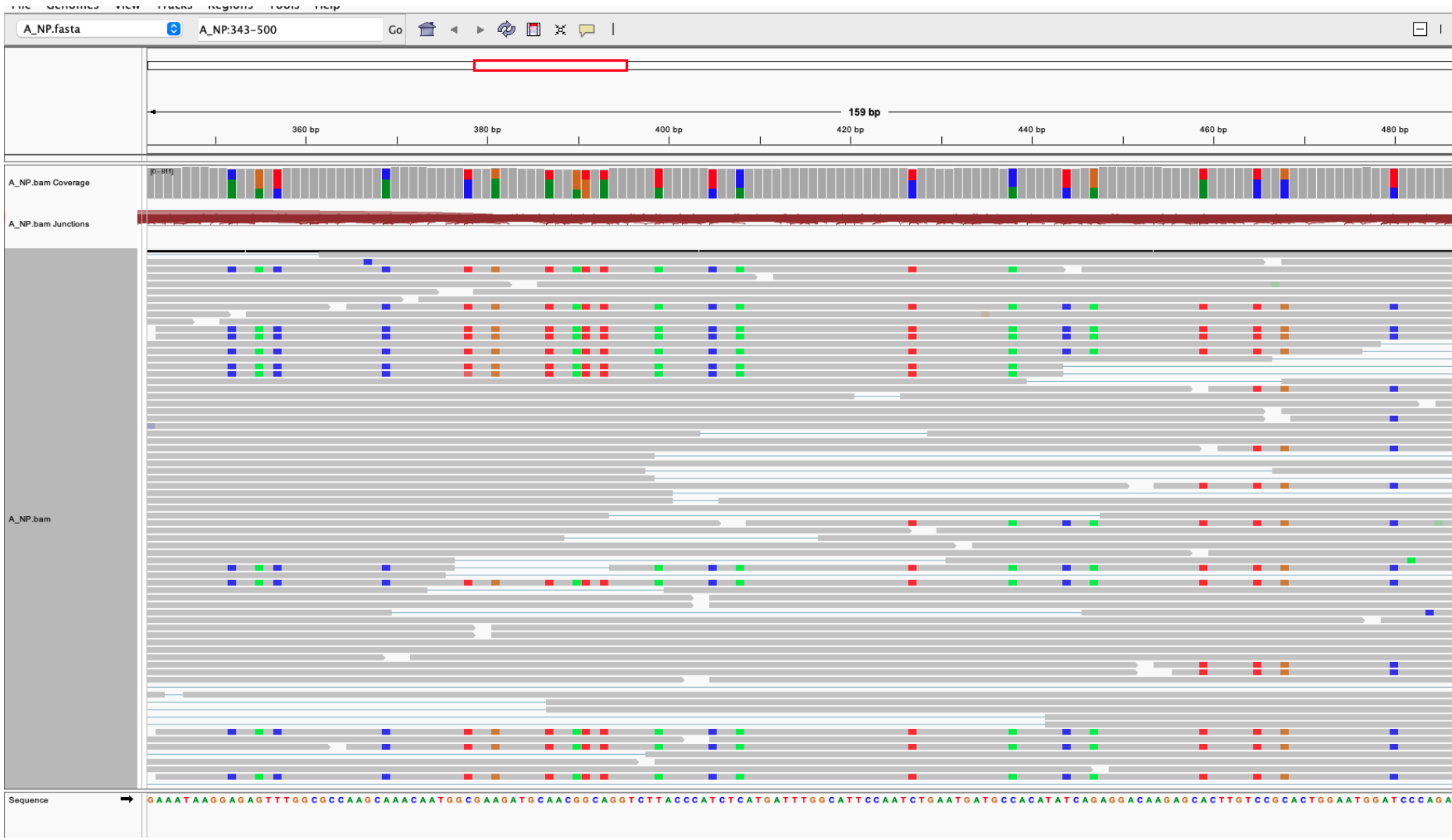
- Failed Mira QC
  - Minor variant count >10
    - In standard *clinical* samples, we have never observed >6 minor alleles (>=5% frequency) with the majority having 1-3, per segment
    - Cell cultures regularly have high genetic variability which can result in high counts
    - Could be a co-infection. Unlucky person got both H1N1 and H3N2 infection *at the same time*
    - **Most likely contamination!**

sample_id	total_reads	pass_qc	reads_mapped	reference	percent_reference_coverage	median_coverage	count_minor_snv_at_or_over_5_pct	pass_fail_reason
95f48e8a	95828	95828	10388	A_HA_H1	99.82	735	2	Pass
95f48e8a	95828	95828	5818	A_MP	100	686	148	Count of minor variants at or over 5% > 10
95f48e8a	95828	95828	9528	A_NA_N1	99.79	814	3	Pass
95f48e8a	95828	95828	9146	A_NP	100	722	274	Count of minor variants at or over 5% > 10
95f48e8a	95828	95828	5704	A_NS	97.1	777	147	Count of minor variants at or over 5% > 10
95f48e8a	95828	95828	11790	A_PA	100	607	361	Count of minor variants at or over 5% > 10
95f48e8a	95828	95828	13230	A_PB1	100	687	246	Count of minor variants at or over 5% > 10
95f48e8a	95828	95828	12175	A_PB2	100	590	391	Count of minor variants at or over 5% > 10
c282a097	16170	16170	1992	A_HA_H3	99.82	158	35	Count of minor variants at or over 5% > 10
c282a097	16170	16170	1382	A_MP	100	187	12	Count of minor variants at or over 5% > 10
c282a097	16170	16170	1852	A_NA_N2	100	178	25	Count of minor variants at or over 5% > 10
c282a097	16170	16170	1720	A_NP	100	157	18	Count of minor variants at or over 5% > 10
c282a097	16170	16170	1336	A_NS	97.1	206	10	Pass
c282a097	16170	16170	2516	A_PA	100	148	22	Count of minor variants at or over 5% > 10
c282a097	16170	16170	2712	A_PB1	100	161	23	Count of minor variants at or over 5% > 10
c282a097	16170	16170	2660	A_PB2	100	160	28	Count of minor variants at or over 5% > 10

Two populations are present.

One matching reference (gray) one sharing all the colored SNVs

2+ mutations on one molecule are “phased” or “linked”

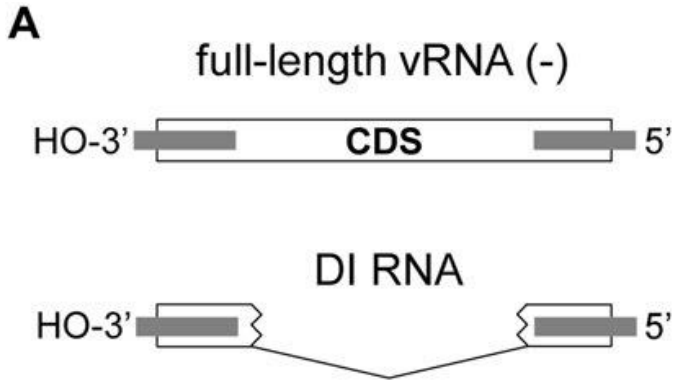


Common issues

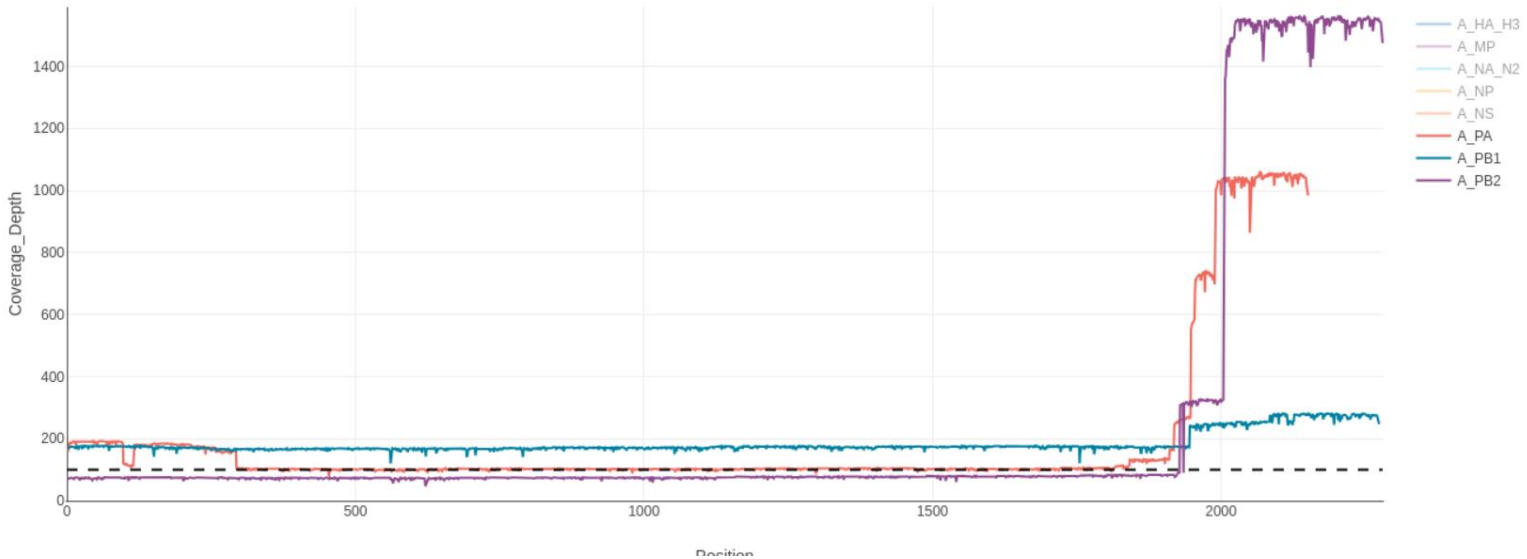
- Failed Mira QC
  - Premature stop-codon?
    - ONT homopolymer issue may require manual correction
    - DI particle induced alignment

DI Particles

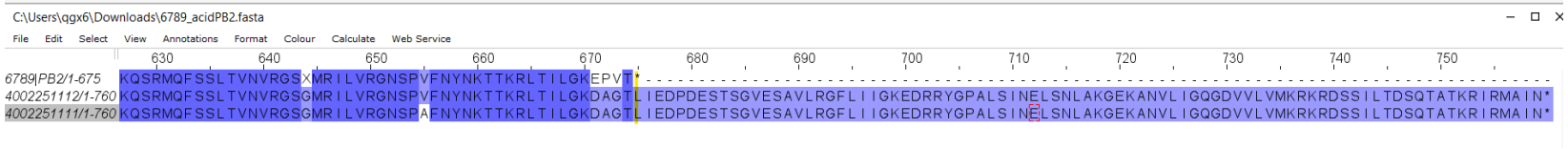
- DI Particles can interfere with NGS
- Common in polymerase segments
- Coverage shape can spike at end or “bat ears”
- Can create erroneous indel mutations at coverage dropoff points



https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/biot.201400429

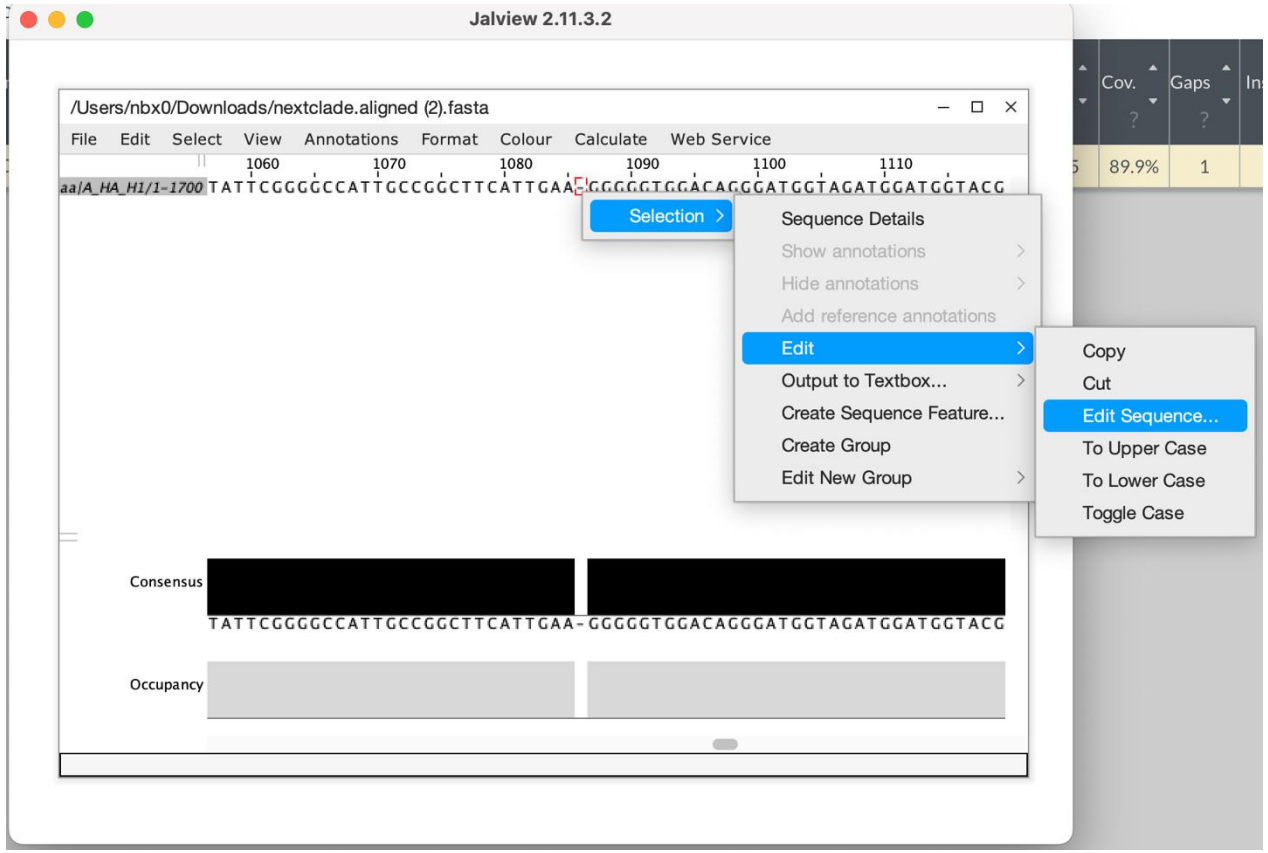


DI Particles



Frameshifts

- Prevalent in homopolymer regions of Oxford Nanopore Sequencing
- DAIS-Ribosome is frameshift-tolerant
  - Shows as (~) mutation
- Convert back to nucleotide space and add or remove base to fix



Mira demands high quality results

- Together we can stop the “Garbage In” side of the data analysis mantra: “Garbage In, Garbage Out”
- Built-in thresholds are for standardizing QC
- Amended consensus gives us extra information from a single sequence.
  - A mixed site may be under active or balancing selection in the host
- Consider each sample as a whole: if HA and NA pass but all other segments fail, consider why.
- High priority samples can be useful even when QC thresholds are not met

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